

ALGEBRAIC PROPERTIES OF COMPLEX NUMBERS

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2.3 Basic Algebraic Properties • Level 1: The Closure Property

Concept: The properties of addition and multiplication of complex numbers are the same as for real numbers. Let z_1 and z_2 be any two complex numbers.

“Do complex numbers obey the same laws as real numbers?”



CLOSURE — ADDITION

$$z_1 + z_2$$

is also a complex number

For any $z_1, z_2 \in \mathbb{C}$

CLOSURE — MULTIPLICATION

$$z_1 \cdot z_2$$

is also a complex number

For any $z_1, z_2 \in \mathbb{C}$

i CLOSURE PROPERTY

The sum and the product of any two complex numbers always belong to the set of complex numbers $\mathbb{C}$.

✦ KEY TAKEAWAY

Complex numbers form a closed system under **addition and multiplication** — the foundation for every other algebraic property.

PROPERTIES UNDER ADDITION

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2.3 Basic Algebraic Properties • Level 2: Commutative, Associative, Identity & Inverse

$$z_1 + z_2 = z_2 + z_1$$

the commutative property

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

the associative property



CONCEPT

Because every real number is also a complex number, the familiar addition laws extend naturally: order and grouping never change the value of a sum of complex numbers.

IDENTITY & INVERSE

Additive identity: $z + 0 = z$, where $0 = 0 + 0i$

Additive inverse: $z + (-z) = 0$

🌟 KEY TAKEAWAY

There exists a unique complex number $0 = 0 + 0i$, and every complex number z has a unique additive inverse $-z$, such that $z + (-z) = 0$.

ADDITION AT A GLANCE

Closure: $z_1 + z_2 \in \mathbb{C}$

Identity leaves z unchanged; inverse cancels it out.

PROPERTIES UNDER MULTIPLICATION

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2.3 Basic Algebraic Properties • Level 3: Rules + Worked Proof



Rule: $z_1 z_2 = z_2 z_1$



WORKED PROOF

Commutative property under addition

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, $x_1, x_2, y_1, y_2 \in \mathbb{R}$

$$\begin{aligned} z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2) \\ &= (x_2 + x_1) + i(y_2 + y_1) \quad (\text{reals commute}) \\ &= (x_2 + iy_2) + (x_1 + iy_1) \\ &= z_2 + z_1 \quad \checkmark \end{aligned}$$

KEY FORMULAS

Mult. identity

$$z \cdot 1 = z$$

Mult. inverse

$$z \cdot w = 1 \quad (z \neq 0)$$

Distributive

$$\begin{aligned} z_1(z_2 + z_3) \\ &= z_1 z_2 + z_1 z_3 \end{aligned}$$

✓ DISTRIBUTIVE PROPERTY

Multiplication distributes over addition, exactly as with real numbers: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ and $(z_1 + z_2)z_3 = z_1 z_3 + z_2 z_3$.

PROOF: THE MULTIPLICATIVE INVERSE

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2.3 Basic Algebraic Properties • Level 4: Key Result, Derivation & Caution

IMPORTANT RESULT

For $z = x+iy \neq 0$,

$$z^{-1} = \frac{x}{x^2+y^2} - \frac{i \cdot y}{x^2+y^2}$$

This is the unique complex number w such that $z \cdot w = 1$.

⚠ CAUTION

z^{-1} is not defined when $z = 0$, since $x^2+y^2 = 0$ only when $x = y = 0$.



CONCEPT UNLOCKED

Every nonzero complex number now has a multiplicative inverse.

PROOF — DERIVATION

Let $w = u+iv$ be the inverse of $z = x+iy$, so $zw = 1$.

$$(x+iy)(u+iv) = 1$$

$$(xu-yv) + i(xv+uy) = 1 + 0i$$

Equating real and imaginary parts:

$$xu - yv = 1 \quad \text{and} \quad xv + uy = 0$$

Solving this system for u and v :

$$u = \frac{x}{x^2+y^2} \quad \text{and} \quad v = -\frac{y}{x^2+y^2}$$

$$(z \neq 0 \Rightarrow x^2+y^2 \neq 0)$$

Therefore:

$$z^{-1} = \frac{x}{x^2+y^2} - \frac{i \cdot y}{x^2+y^2} \quad \checkmark$$